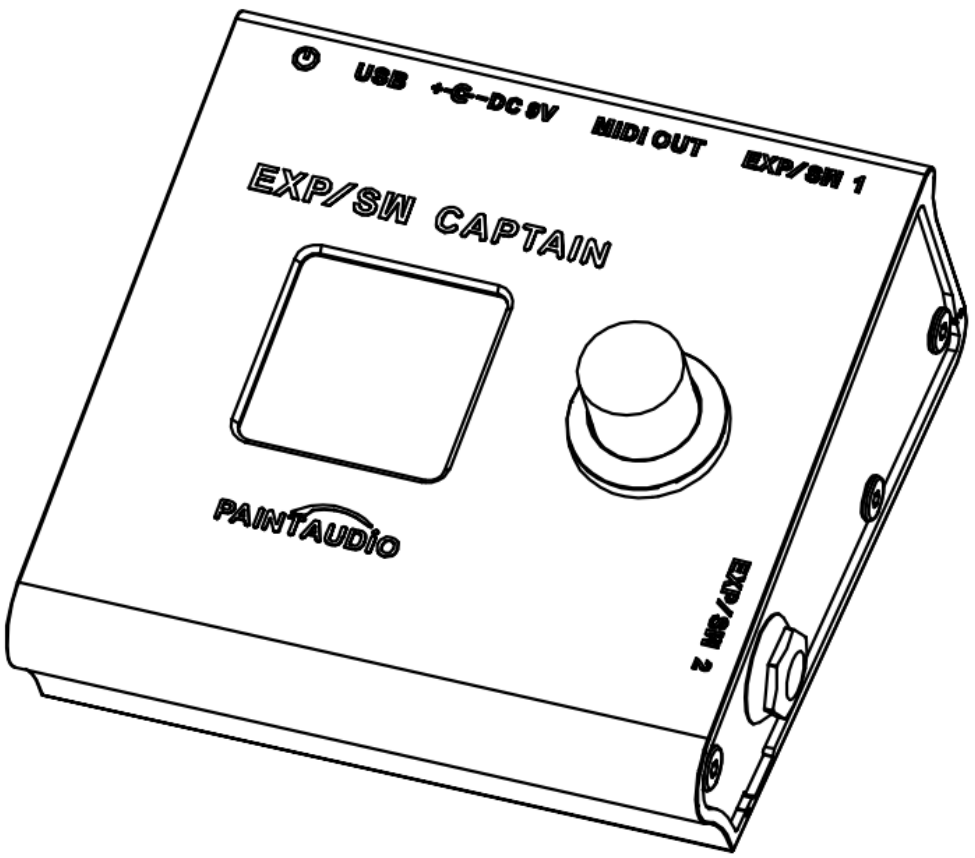
EXP/SW CAPTAIN

Expression/Switch PC/CC MIDI Command Controller

Unique Expression and Switch Linkage Control Function

It can support up to the following simultaneously:

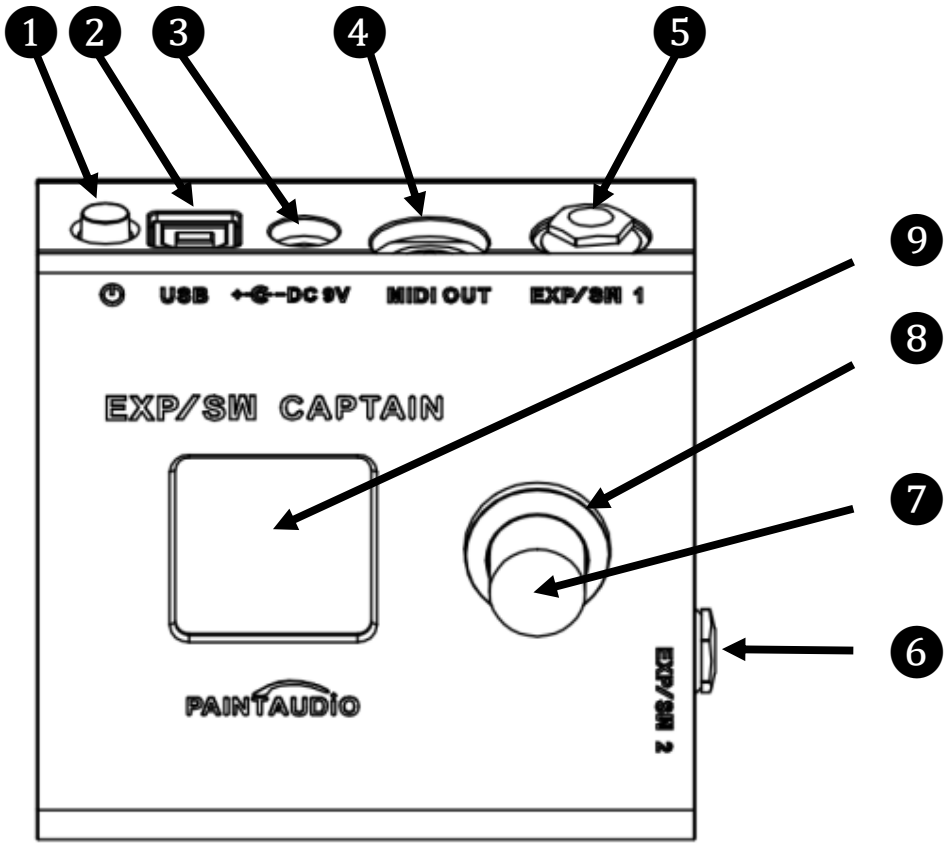
- 2 channels of expression pedal input (Mode A/Mode B)
- 4 channels of switch input (SWL/SWM)
- 1 channel of expression and 2 channels of switches
- Supports setting the upper and lower value limits for the expression pedal



PAINTAUDIO

This manual is continuously updated along with the product and is for reference only. Ver1.0 (2025-8)

1 Interfaces and Controls



1.Power Switch

2.USB Interface (supports USB power supply and USB MIDI)

3.DC Power Interface (supports 9-18V DC power supply, compatible with different polarities)

4.MIDI Signal Output Port

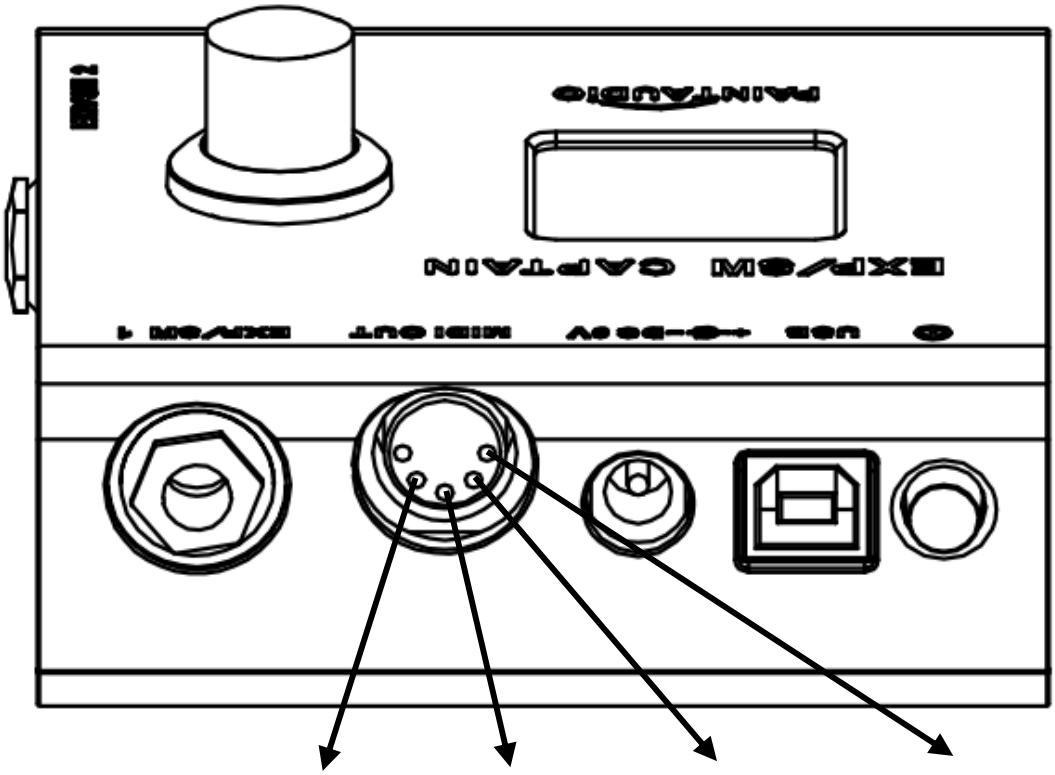
5.Expression Pedal & Switch Input Channel 1 (EXP supports Mode A and Mode B; SW supports SWL and SWM modes)

6.Expression Pedal & Switch Input Channel 2 (EXP supports Mode A and Mode B; SW supports SWL and SWM modes)

7.Encoder Knob with Buttons

8.Status Indicator Light Ring

9.1.54-inch 240×240 Pixels Color Screen

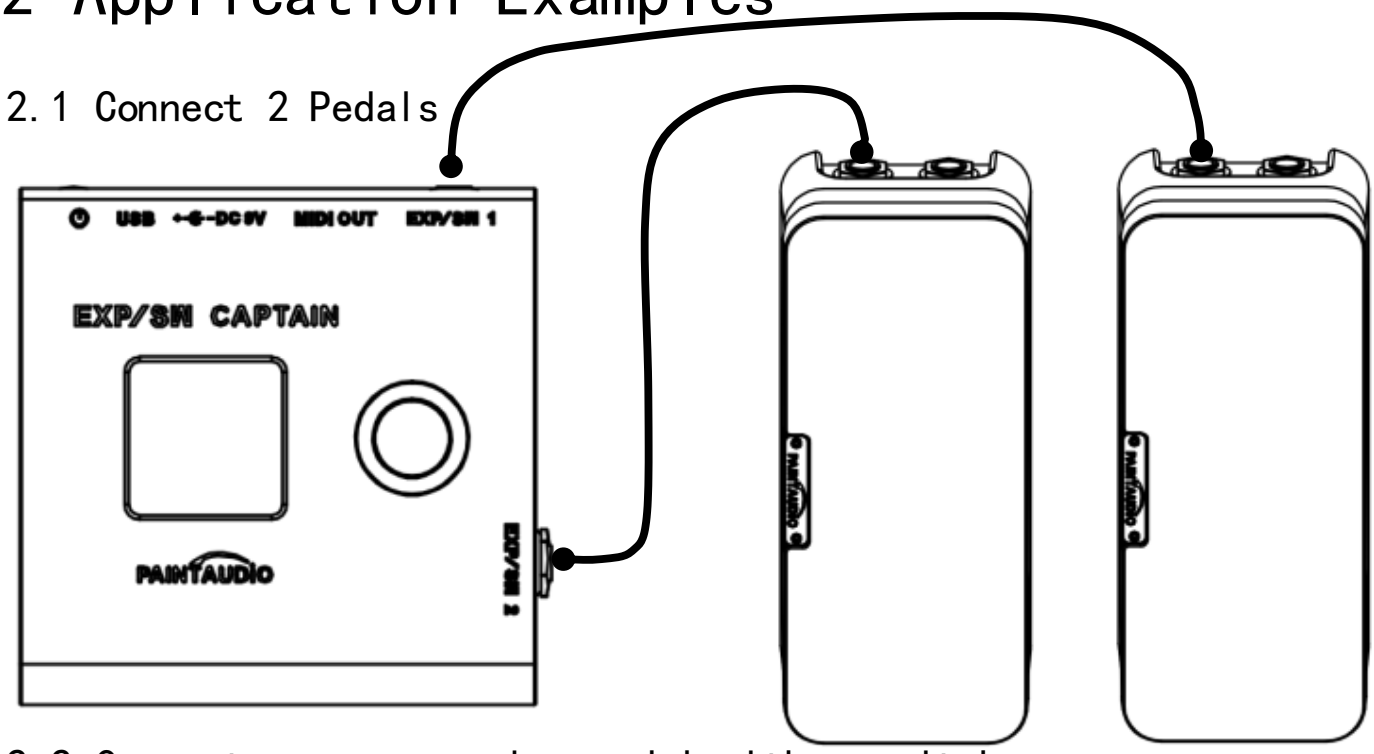


・This product can be directly powered by devices with MIDI auxiliary power supply via a full 5-PIN MIDI cable (it can be directly powered via the MIDI port by all series of MIDI CAPTAIN).

Special note: Since some devices have pin 1 grounded and some other devices also have MIDI auxiliary power supply function, for better compatibility, all PAINTAUDIO products are equipped with 3-PIN MIDI cables, which only support MIDI communication connections. If you need to use a device with auxiliary power supply to directly power this product, you need to purchase an additional full 5PIN MIDI cable.

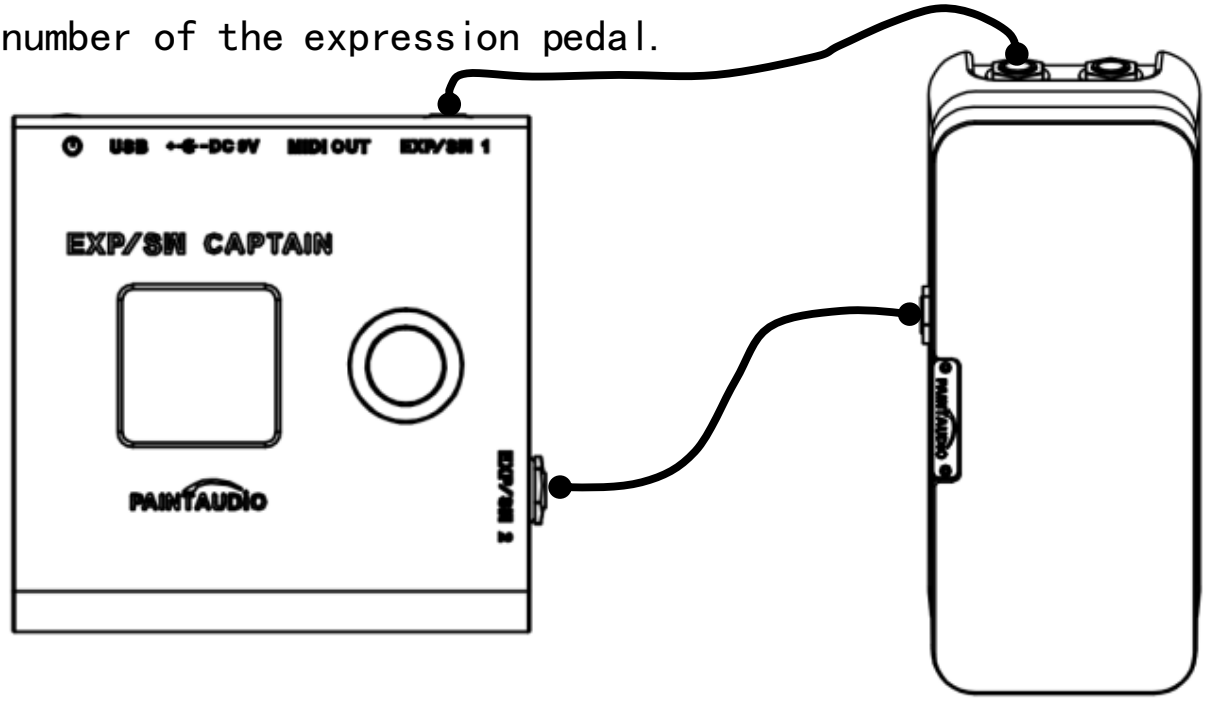
2 Application Examples

2.1 Connect 2 Pedals

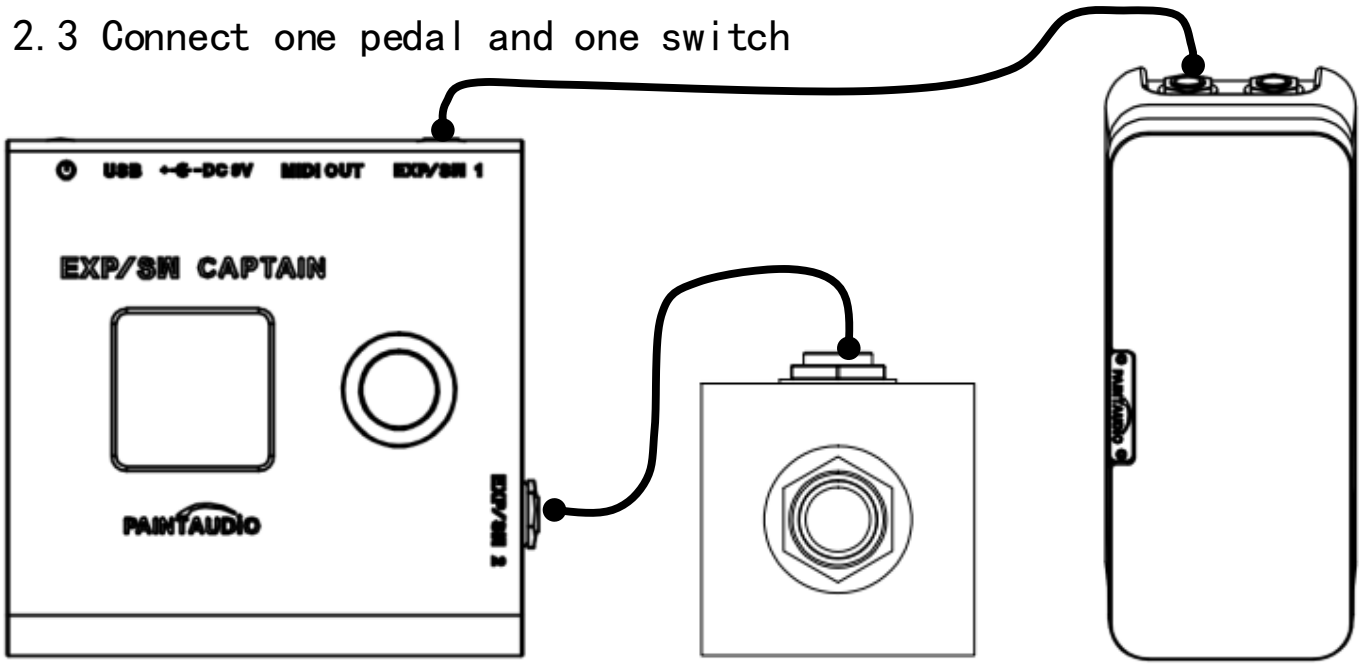


2.2 Connect one expression pedal with a switch.

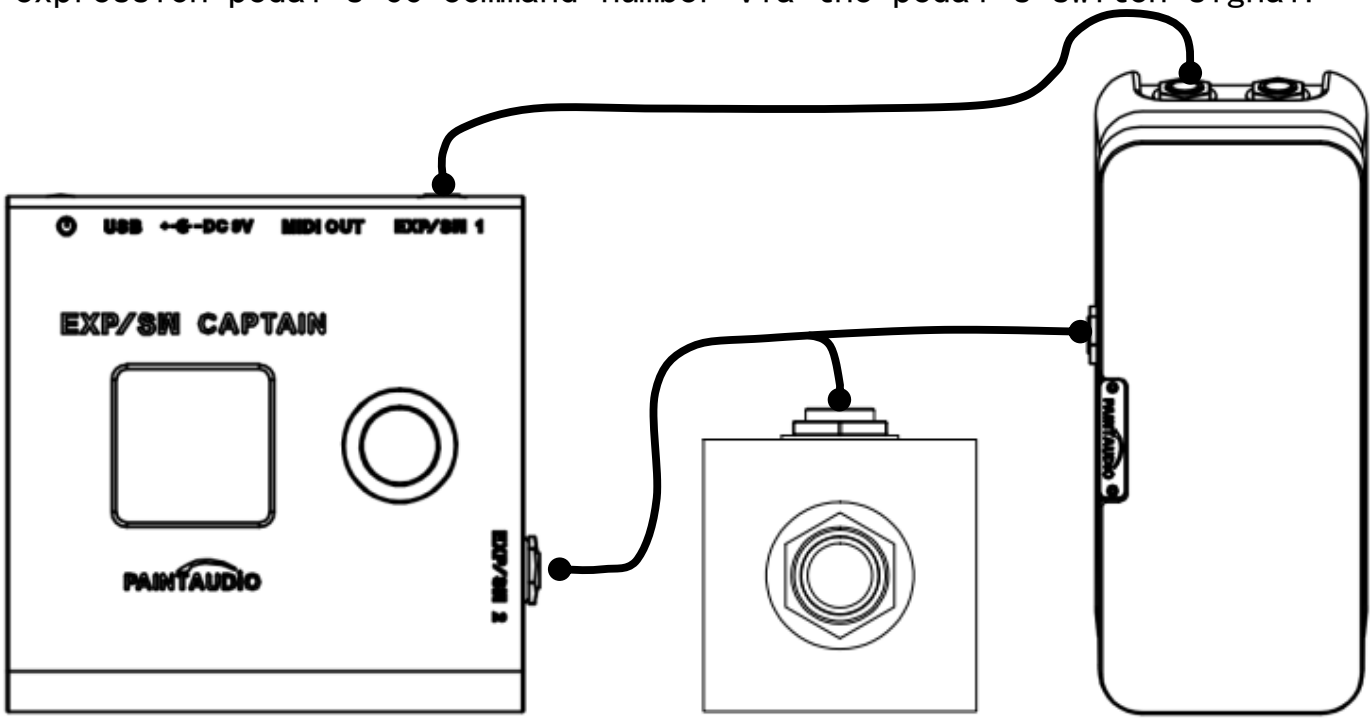
It supports sending PC/CC commands via the switch and enables linkage triggering through the switch to switch the CC command number of the expression pedal.



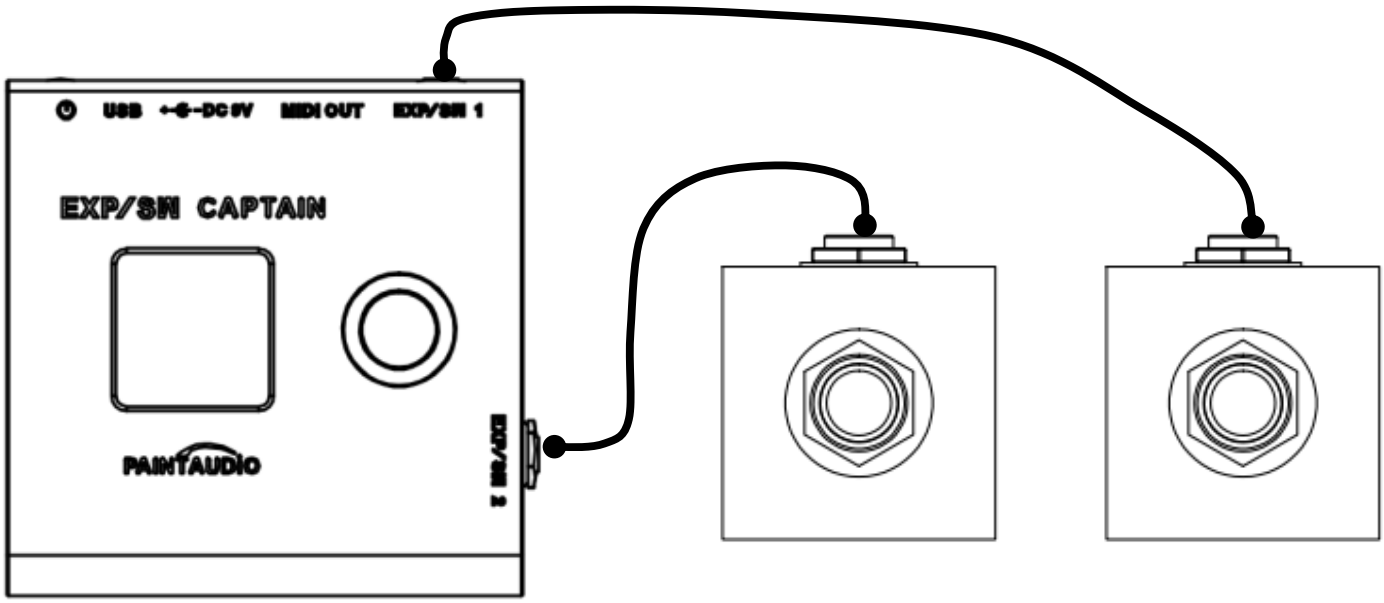
2.3 Connect one pedal and one switch



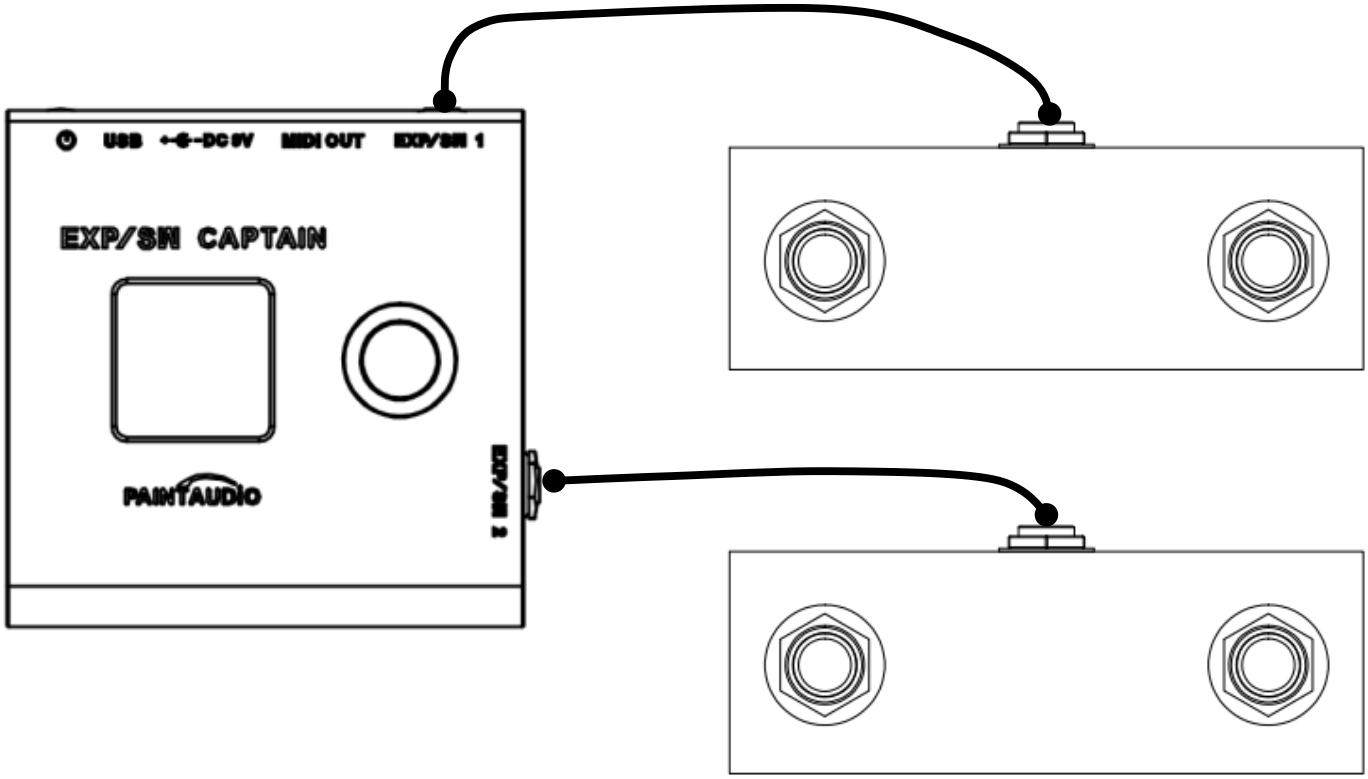
2.4 Connect one expression pedal with a switch, and connect one 1-pole switch at the same time. Use a 6.35mm stereo-to-mono cable. Note that the left channel interface must be connected to the switch port of the expression pedal to enable linkage switching of the expression pedal's CC command number via the pedal's switch signal.



2.5 Connect 2 1-pole switches

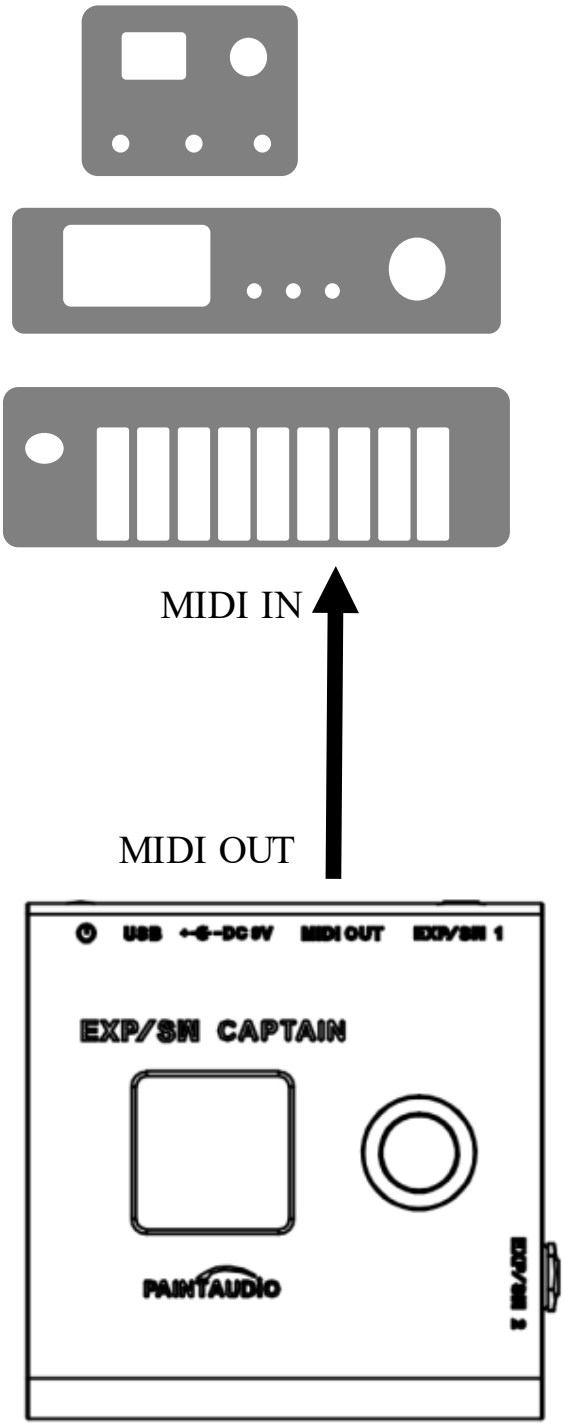


2.6 Connect 2 2-pole switches



3 Connection Method

3.1 Traditional MIDI Communication Connection



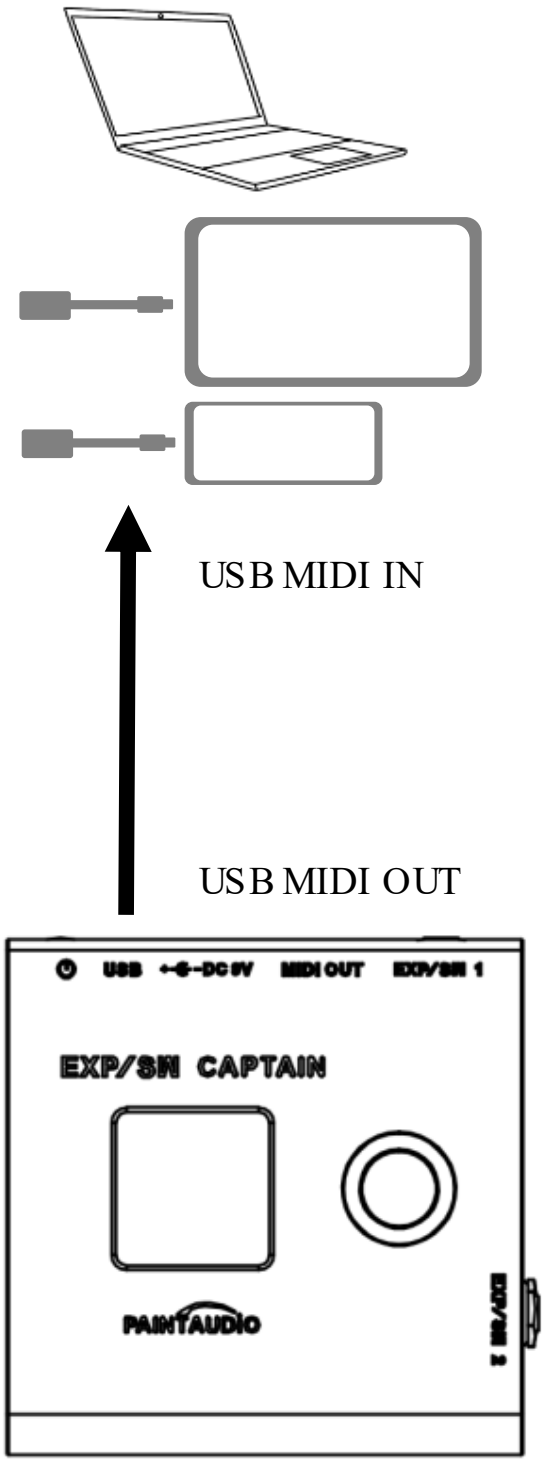
It can control most hardware devices with a MIDI IN interface, including but not limited to various desktop multi-effects processors, speakers, keyboards, synthesizers, and sound source devices.

- Line6 HX Stomp
- Boss GT1000Core(need convert cable)
- Fractal AXE FX2
- Fractal AXE FX3
- Fractal FM3
- Eventide H9
- Kemper Profiler Power Head
- Kemper Profiler Rack
- Hotone Ampero one Stomp
- Hotone Ampero one II Stomp
  FLAMMA FX200
- MOOER GE250
- HeadRush Gigboard
- HeadRush MX5 (need convert cable)
- LINE6 POD HD PRO X

Most keyboards and digital pianos

… …

3.2 Software and Connect with USB Host MIDI Communication

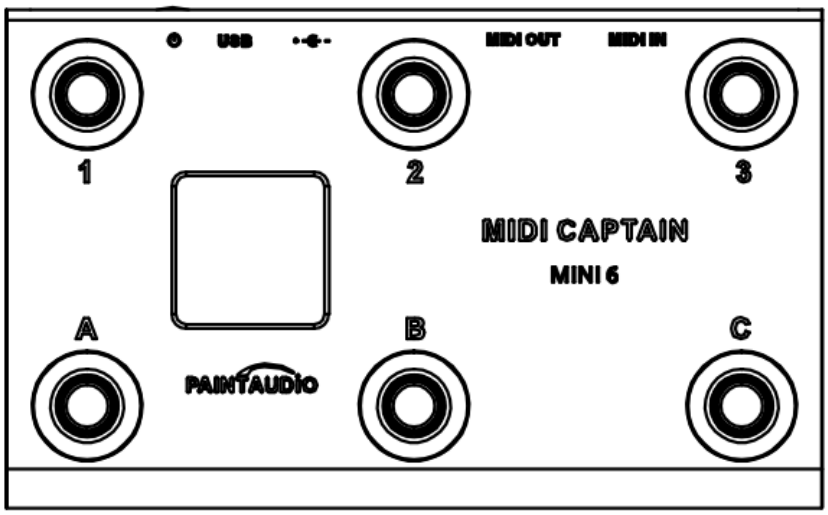


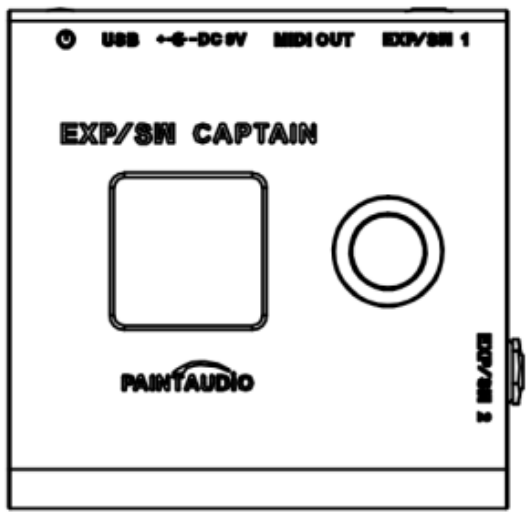
It is compatible with nearly all professional audio software, effects software, and recording software on computers, tablets, and mobile phones via USB-MIDI adaptation.

Most software requires selecting "PaintAudio MIDI" as the software's MIDI signal input source in the software after the device is connected to the computer.

Note: This device is a USB slave device and can only communicate with USB host devices such as computers and tablets; it cannot communicate with other USB slave devices.

3.3 Cascade connection via other MIDI devices (the devices must have the MIDI Through function)

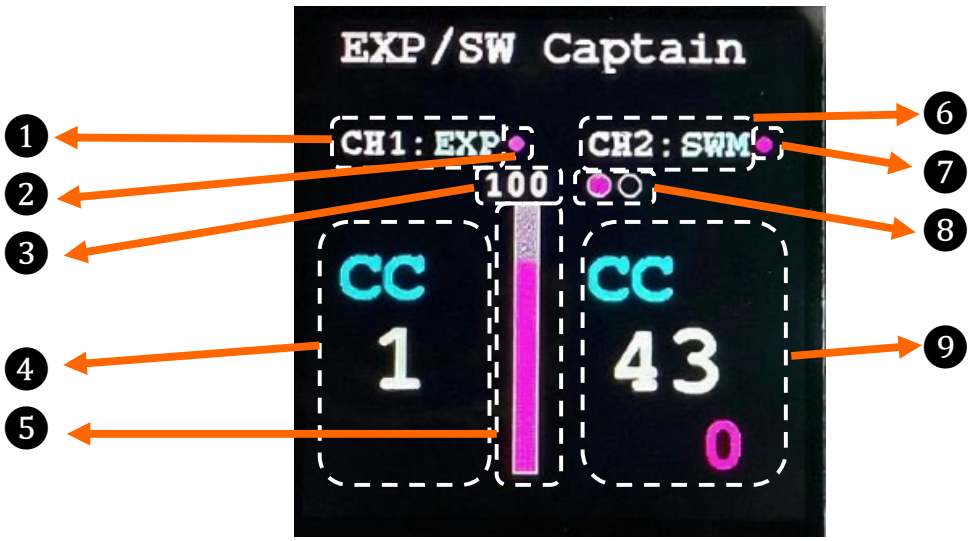




EXP/SW CAPTAIN can perform cascade communication with MIDI CAPTAIN BLUE, GOLD, MINI6 and NANO4 via a MIDI cable. Simply connect the MIDI OUT port of EXP/SW CAPTAIN to the MIDI IN port of MIDI CAPTAIN, and enable the MIDI Through function of MIDI CAPTAIN to use them in series.

Use a full 5PIN MIDI cable (the MIDI cable provided with the device is a 3PIN cable; a full 5PIN cable needs to be purchased separately). MIDI CAPTAIN can directly power EXP/SW CAPTAIN without the need for an additional power supply.

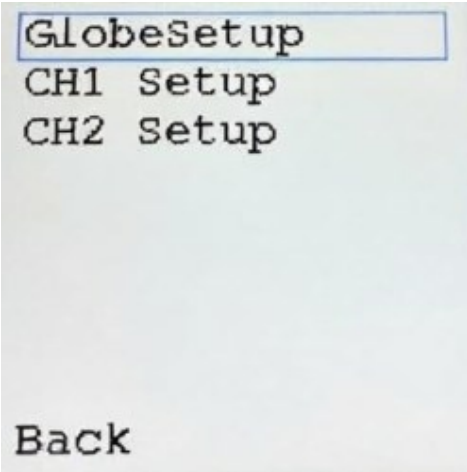
4 Screen Display



1. Channel 1 input mode indicator, with three modes: EXP, SWL, and SWM.
2. Channel 1 insertion indicator, showing a purple dot when a device is inserted, and empty when no device is inserted.
3. Expression pedal input value
4. Transient display of MIDI commands sent by Channel 1
5. Expression pedal control status display
6. Channel 2 input mode indicator, with three modes: EXP, SWL, and SWM
7. Channel 2 insertion indicator
8. Display of cycle count and current status of Channel 2 switch
9. Transient display of MIDI commands sent by Channel 2

The content and status displayed on the channel section of the screen are related to the currently selected input mode. When the channel mode is set to EXP, the displayed status is the same as that of Channel 1 in the example diagram; when the channel mode is set to SWL or SWM, the displayed status is the same as that of Channel 2 in the example diagram.

4 Quick Operation Guide



1、Press and hold the right encoder knob to enter the settings interface.

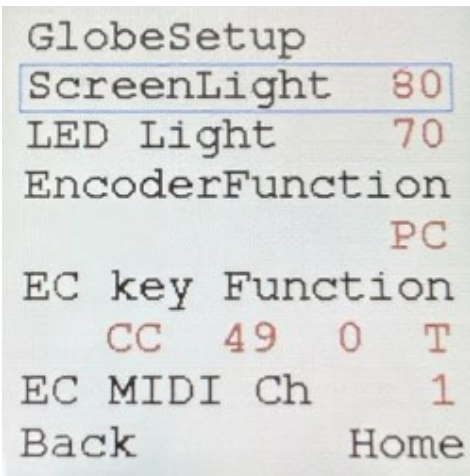
2、Rotate to select and enter the next interface:

GlobeSetup: Device Global Settings ;

CH1 Setup: Channel 1 Settings ;

CH2 Setup: Channel 2 Settings ;

Back: Return to the main interface 。



Global Settings Interface

1. ScreenLight: Screen backlight brightness.
2. LED Light: LED light brightness.
3. EncoderFunction: Encoder function setting .

None: None.

PC: rotate the encoder to send PC commands.

CC: rotate the encoder to send CC commands.

4. EC Key Function:

None: None.

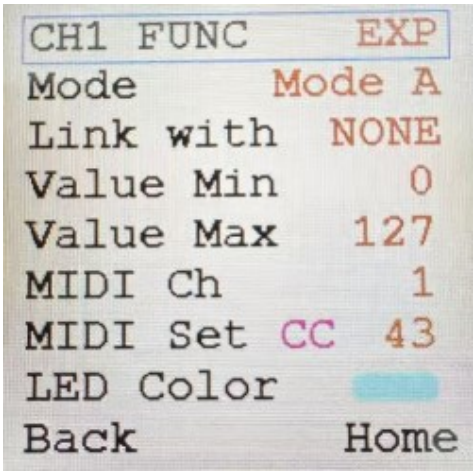
PC: press the key to send the set PC command

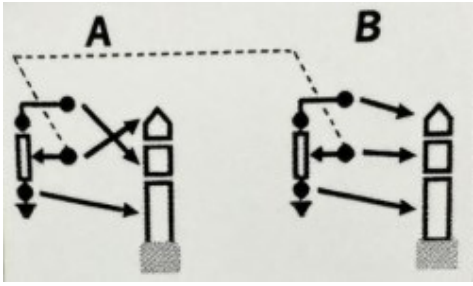
CC: Return to the main interface, press the key to send the set CC command. When "T" is selected, the command toggles with the number of presses.

5. EC MIDI Ch: MIDI command channel setting for the encoder, with a range of 0-16 .

Channel Settings Interface:

The channel is set to "Expression Pedal" and does not link with the switch input of the other channel.





Wiring Diagram for Mode A and Mode B

1. CH1 FUNC: Channel 1 function selection. Set to EXP function
2. Mode: Select based on the type of expression pedal used, with two modes: Mode A and Mode B.
3. Link with: Whether the expression pedal is controlled in linkage with the other channel. Selection: NONE
4. Value Min: Lower limit of the expression pedal value.
5. Value Max: Upper limit of the expression pedal value.
6. MIDI Ch: MIDI signal output channel.
7. MIDI Set: Setting for the CC command.
8. LED Color: Selection of LED status

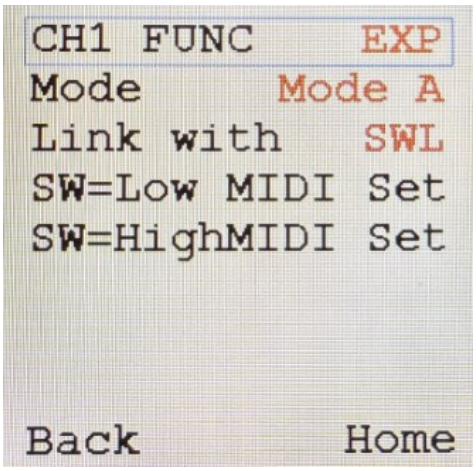
Note:

1. An TRS cable is required for connecting the expression pedal to the device.

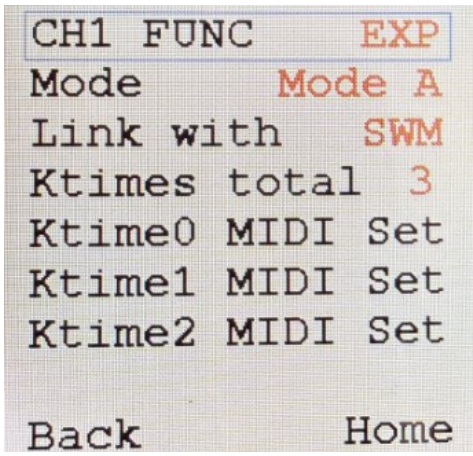
2. If you are unsure whether the expression pedal you are using adopts Mode A or Mode B, you can first select Mode A, then return to the main interface, step on the expression pedal continuously, and check if the sampling value displayed on the device changes continuously and evenly. If the sampling value changes continuously and evenly, your selection of the expression pedal type is correct. If not, enter the settings page and set the pedal mode to Mode B.

Channel Settings Interface:

The channel is set to "Expression Pedal" and links with the switch input of the other channel.



Link with SWL



Link with SWM

1. Link with: Select switch parameters of the same type as the switch inserted in the other channel. SWL is a latching switch ,and SWM is a momentary switch.

2. Link with SWL : Enter the options of SW=Low (switch closed) and SW=High (switch open), and set the specific parameters of the expression pedal's CC commands.

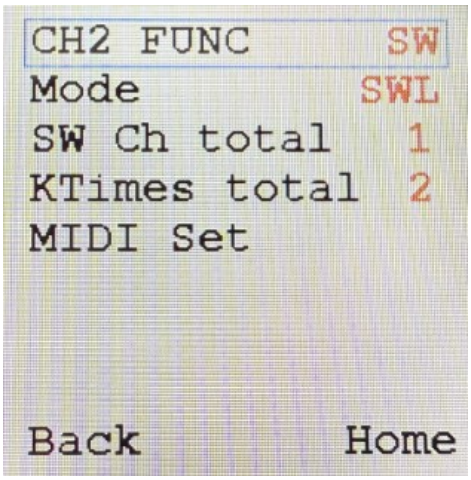
3. Link with SWM : Enter Ktimes total (total switch cycle count) to set the maximum number of switch cycles . And Enter Ktime0, Ktime1, Ktime2 set the specific parameters of the expression pedal's CC commands.

Note:

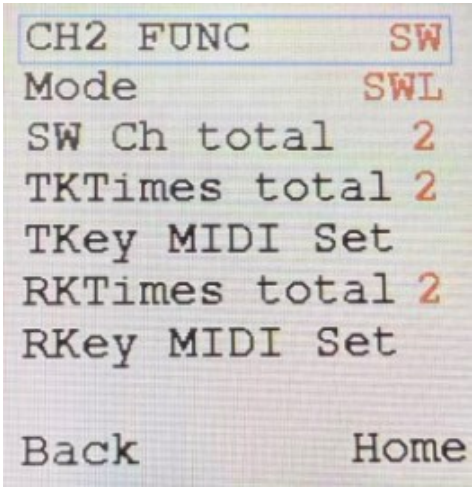
For SWL switche, the expression pedal only supports linked changes with the switch state when the switch is in the two states of **on** and **off**. For example, when SW=L, send CC 01; when SW=H, send CC 02.

For SWM-type switches, the maximum cycle count is 3, and the expression pedal can be set to 3 different linked commands. For example: the CC 01 command is sent at Ktime0, the CC 02 command at Ktime1, and the CC 03 command at Ktime2 .

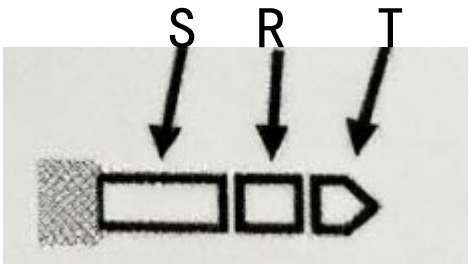
Channel Settings Interface: The channel is set to SWL



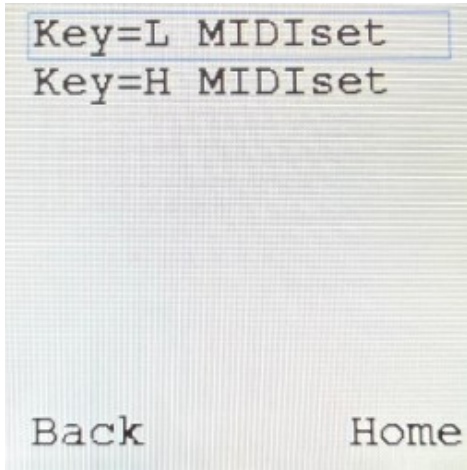
Set to SW & CH = 1



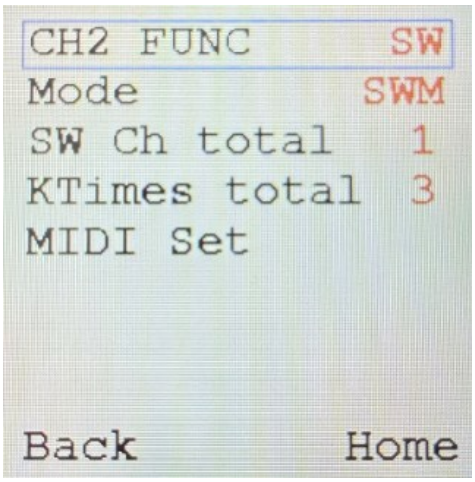
Set to SW & CH = 2



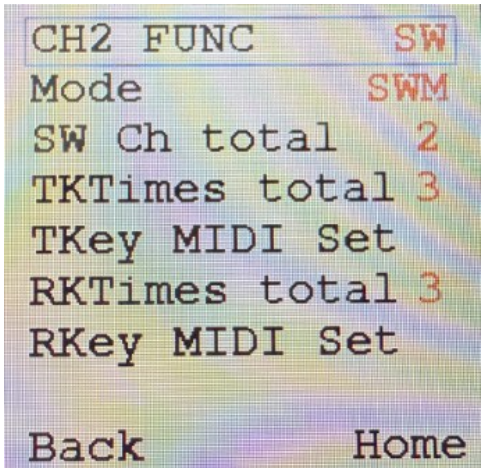
1. CH FUNC : SWL.
2. When the number of connected switches is 1 (single-pole switch), in this case, the switch is connected to the T segment of the TRS cable . When the number of connected switches is 2 ; in this case, the switches are connected to the T segment and R segment of the TRS cable respectively, and the S segment serves as the common ground for the switches
3. KTimes total: Selection of the switch cycle count. You can select 1 cycle or 2 cycles.
4. When "2 cycles" is selected for KTimes total, the content displayed when entering the MIDI Set interface is as shown in the bottom right corner.



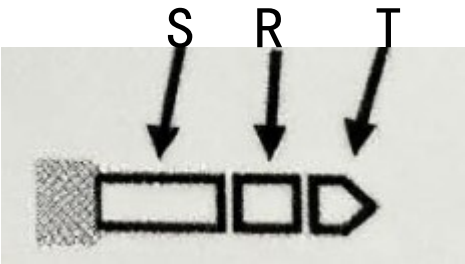
Channel Settings Interface: The channel is set to SWM



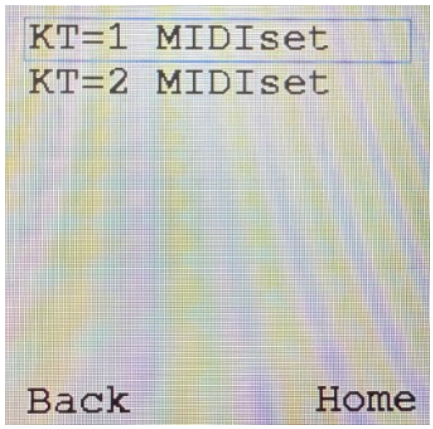
Set to SW & CH = 1



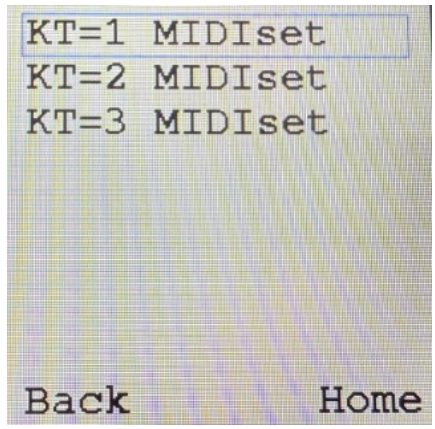
Set to SW & CH = 2



1. CH FUNC : SWM.
2. When the number of connected switches is 1, in this case, the switch is connected to the T segment of the TRS cable by default. When the number of connected switches is 2, in this case, the switches are connected to the T segment and R segment of the TRS cable respectively, and the S segment serves as the common ground for the switches.
3. KTimes total: Selection of the switch cycle count. You can select 1 cycle, 2 cycles, or 3 cycles.
4. When "2 cycles" or "3 cycles" is selected for KTimes total, the content displayed when entering the MIDI Set interface is as shown.

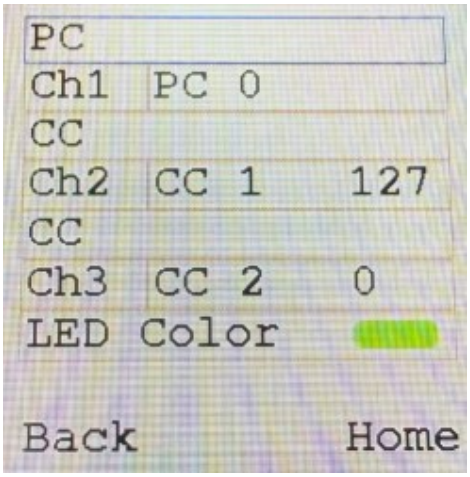


KTimes total = 2



KTimes total = 3

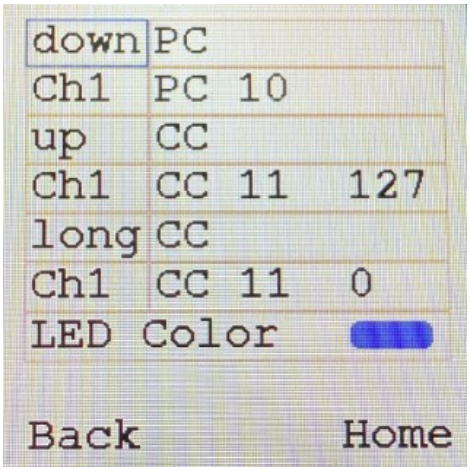
MIDI指令设置界面



SWL MIDI Set

Description of the MIDI Command Settings Interface When the Channel is Set to SWL

1. Each switch state supports sending a maximum of 3 groups of MIDI commands simultaneously.
2. Support: PC、CC、PC Inc、PC Dec、CC Value Inc、CC Value Dec。
3. Each command can have its MIDI command channel set independently.
4. LED Color：Setting for the color of the LED.



SWM MIDI Set

Description of the MIDI Command Settings Interface When the Channel is Set to SWM

1. Each switch state supports sending a maximum of 3 groups of MIDI commands simultaneously.
2. Support: PC、CC、PC Inc、PC Dec、CC Value Inc、CC Value Dec。
3. You can set the command transmission status via down (transmit on press), up (transmit on release), and long (transmit on long press).

Note:The long command and the up command are mutually exclusive. When the long command takes effect, the up command will no longer be transmitted when the switch is released.

4. LED Color：Setting for the color of the LED.

5 Explanation of MIDI Command Functions

When the device uses the switch command function, it provides a total of 6 command types, namely PC, CC, PC Inc, PC Dec, CC Value Inc, and CC Value Dec.

1. PC: When a key is triggered, the set PC command is sent.
2. CC: When a key is triggered, the set CC command is sent.
3. PC Inc: Automatically increments the PC command count by 1.
4. PC Dec: Automatically decrements the PC command count by 1
5. CC Value Inc: Automatically increments the CC Value command count by 1. For example, if the current automatically counted command is CC 43 00 command, after the key is triggered, the CC 43 01 command is sent.
6. CC Value Dec: Automatically decrements the CC Value command count by 1. For example, if the current automatically counted command is CC 43 01 command, after the key is triggered, the CC 43 00 command is sent.

Note: The automatically counted PC and CC Value values in PC Inc, PC Dec, CC Value Inc, and CC Value Dec are completely independent of the PC commands and CC commands.

For example, if the PC command is set to send the "PC 8" command, the count of the automatically counted PC command remains 0. After sending the "PC 8" command, when the PC Inc command is executed again, the sent command is still the "PC 1" command .

6 technical indicators

Supply Voltage	5V (USB) ; 9-15V (DC in)
Supply Current	0.1A - 0.4A (Maximum Brightness of LED and Screen)
Power	0.5W - 2W (Maximum Brightness of LED and Screen)
Screen	240x240 RGB 24bits LCD
MIDI Interface	MIDI OUT(supports power supply via the MIDI interface)
Interface	2 x 6.35mm jacks
Weight	
Dimensions	L x W x H: 99mm x 97mm x 51mm

7 Contact Us

Email:

contact@paintaudio.com

support@paintaudio.com

Web: www.paintaudio.com

8 Safety Information

Always follow the basic precautions listed below to avoid the risk of serious injury or even death caused by electric shock, damage, fire, or other hazards. These precautions include but are not limited to the following:

- Do not connect or use the device during a thunderstorm.
- Do not install power cords or sockets in damp areas.
- If the device requires AC power supply, do not touch the exposed parts of the power cord or the connectors when the power cord is connected to an AC power socket.
- When setting up the device, carefully follow the instructions in the manual.
- Do not expose the device to rain or humid environments to prevent fire or electric shock.
- Keep the device away from power appliance interfaces, such as electric motors.
- Disconnect the USB cable and the adapter power connection when the device is not in use.
- Do not touch the device with wet hands.

9 Other Statements

This manual is provided solely as a reference for the standardized use of the product. Any method of use not specified in this instruction manual shall be deemed improper use of the product. PaintAudio shall not be liable for any losses caused by non-standard or improper use, as well as other human errors. This manual may contain relevant information such as names, keywords, or patterns of other products and software. Such references are used solely for supplementary explanation and reference purposes. If registered trademarks or patents are involved, the relevant owners may submit applications for modification or deletion. PaintAudio reserves the right to modify, add to, or delete content in this document due to product upgrades and replacements.

保修卡 Warranty

型号 Model	
用户签名 Signature	
日期 Date of Sales	
地址 Address	
序列号 SN	
经销商 Distributor	
发票号 Invoice NO.	
其他备注 Others	

本产品自购买之日12个月内享受免费维保服务。以下情形除外：不可抗力损坏；人为损坏；私自拆修；法规约定的其它情形。

This product enjoys free maintenance service within 12 months from the date of purchase. Except for the following situations: force majeure damage; man-made damage; unauthorized dismantling and repair; other circumstances stipulated by laws and regulations.

PAINTAUDIO

WARRANTY 1 YEAR WARRANTY

合　　格　　证

型号　MODEL	
检验　CHECKER	
日期　DATE	

PAINTAUDiO

THANK YOU